

FALCON's Response to Delhivery for RFQ of Bag Sorter_Bangalore

May 13, 2025





Kind Attention –

Mr. Sunny Raja

Mr. Aman Goel

M/s DELHIVERY

Offer Ref: F24-00609-01; Date 01.02.2025.

Subject – Techno -Commercial Offer of Bag Sorter- Phase 2

We are pleased to submit our Techno-Commercial Offer to DELHIVERY in response to your **RFQ for Bag Sorter, Bangalore**. We have taken this opportunity to also include adjustments discussed during DAP till **10th Dec 2024**.

We would like to highlight the fact that Falcon Autotech, as a prime contractor for this project, is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative. From technical point of view, the system is **designed at 9,000 BPH** and ensures simple operations and movement within the facility.

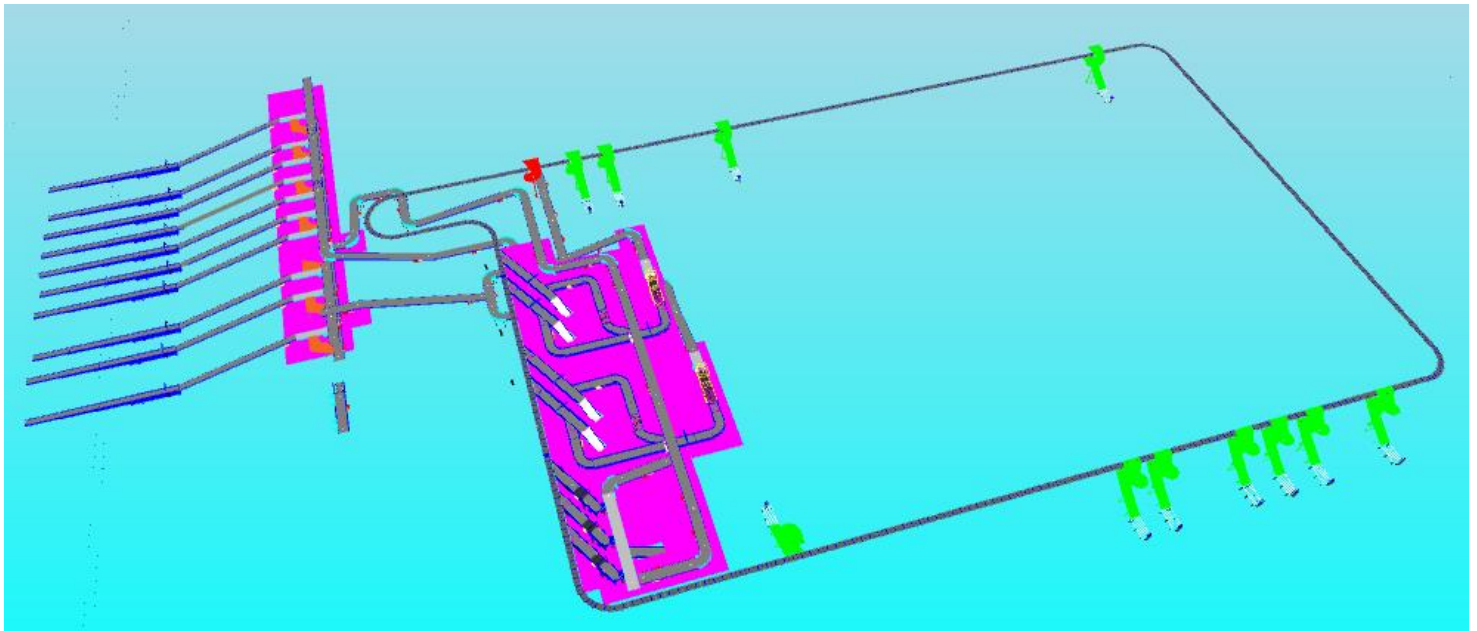
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Rishabh Jain

Vice President- Sales

Response to RFQ for Bag Sorter_ Delhivery, Bangalore



Proposal Reference – F24-00609-01

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Contents

Contents	4
1. Glossary	7
2. Executive Summary	8
3. Company Profile	9
4. Falcon's Experience and Achievements in Sortation Space Globally	13
5. Reference Projects	15
5.1 Project 1- (CEP Client, UK)	15
5.2 Project 2- (CEP Client, Riyadh)	16
5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)	17
5.4 Project 4- (CEP Client, Sydney)	17
5.5 Project 5- (Client – E-Commerce, India)	18
5.6 Project 6- (Ecommerce Client, India)	19
6. Handled Parcel Spectrum	20
6.1 Parcel size loadable on the sorter	20
6.2 Parcels to be loaded on Sorter shall have the following characteristics:	20
6.3 Parcel not loadable on the sorter	21
6.4 Pictures of Sortable Parcels	22
7. Proposed System Description	23
7.1 Objective	23
7.2 Summary of the System	23
7.3 Main benefits of the proposed solution	24
8. Layout View	25
9. Proposed System Capacity Calculations	26
9.1 Sorter System Capacity	26
10. System description	26
10.1 Unloading Area	26
10.2 Infeed System	27
10.2.1 Telescopic Belt Conveyors	27
10.2.2 Infeed Conveyor System	28
10.2.2.1 Straight and Inclined Powered Belt Conveyor	28
10.2.2.2 Plastic Modular Belt Conveyor	30
10.2.2.3 Curve Conveyor	31
10.2.2.4 Buffer Conveyor	31
10.2.2.5 Belt Merge	31
10.2.2.6 Aligning Conveyor	32

10.2.3	Conveyor BOQ	33
10.3	Feedlines	35
10.3.1	Spacing Conveyor	36
10.3.2	Buffer Conveyors	37
10.3.3	Weighing Conveyor	37
10.3.4	Receiving Conveyor	38
10.3.5	Intelligent Merge Conveyor	38
10.4	Main Loop	39
10.5	Sorting Output	40
10.5.1	Staging Chute	40
10.5.2	DBO Chute	41
11.1	Mechanical equipment	42
11.2	Electrical Equipment	43
12.	Description of Components of Equipment	44
12.1	Cross Belt Sorter -	44
12.2	CBS Carrier	44
12.2.1	Servo Roller	45
12.2.2	Chassis	45
12.2.3	Carrier Wheels	45
12.2.4	Friction Wheel Drive	46
12.2.5	Non-contact based linear motor drive	46
12.2.6	Power Transmission	46
12.2.7	Data Transmission	47
12.2.8	Carriers positioning System.	47
12.2.9	Technical specifications of Sorter	48
12.3	Scanning & Sensing on Main Sorter Loop	49
12.3.1	Automatic Barcode Scanning & Dimensioning System	49
12.3.2	Parcel Positioning System.	49
12.3.3	Empty Carrier Detection System	49
13.	Electrical System	50
13.1.1	Reference Picture of Power Distribution Panel	50
13.1.2	Main Control Panel (Reference)	50
13.1.3	Induct Stations Control Panel (Reference)	51
14.	Controls Architecture (Ref.)	52
15.	IT Architecture (Ref.)	52
16.	Key Components Make	53



17.	System Handover to Delhivery Team	54
	User Acceptance Test Parameters	55
18.	Warranty Period	57
19.	Commercial Terms.....	58
19.1	Price Sheet	58
19.2	Payment Terms.....	58
20.	Exclusions	59

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	ARB	Actuated Roller Balls
4	DBO	Damaged Barcode
5	VDS	Volume Distribution System
6	ICR	Intelligent Character Recognition
7	MEZZ	Mezzanine
8	LIM	Linear Induction Motors
9	ECDS	Empty Carrier Detection System
10	AC	Alternating Current
11	DC	Direct Current
12	PLC	Programmable Logic Controller
13	IT	Information Technology
14	BOQ	Bill Of Quantity
15	I/O	Input/ Output
16	PDP	Power Distribution Panel
17	PC	Personal Computer
18	UPS	Uninterrupted Power Supply
19	CBS	Cross Belt Sorter
20	DNF	Data Not Found
21	LGM	Logic Mismatch
22	RTVC	Real Time Video Coding

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this RFQ of Bag Sorter. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.

As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the Bag Sorter system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **One loop of Bag sorter, based on a cross-belt technology, offers a designed throughput of 7000-8000 pph. (Detailed TPH calculation will be shared with Delhivery Team during the DAP stage.)**
- **Infeed conveyor system including 10 telescopic conveyors.**
- **11 Nos staging chutes, 1 No DBO Chute, 7 Nos Bag Transfer chutes on Infeed Line, 4 Nos Irregular Handling Chute on Auto Induct and 1 No Irregular chute on Manual Dump Loop VDS.**

A tailor-made and simple layout, specifically designed to Delhivery The proposed layout, is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our last Teams workshop meeting on June 8th.

- Simple operational conditions due to one single loop cross belt sorter.
- Easy maintenance: optimized number of conveyors and concentrated inducts area.
- Vehicle movement zones.
- Exceptions handling area with direct feeding via conveyor.

1. **Falcon's reliable and fully CE compliant Parcel sortation systems**

These systems are globally being used by most innovative brands such as Aramex, Amazon, Asendia, Fastway, Delhivery and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. **Commitment to quality systems**

Demonstrating Falcon's clear commitment to the Delhivery Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 12 months from the completion of installation & commissioning.

3. **Commitment to ensuring the quality of the delivery and its schedule.**

Falcon proposes a ramp-up approach, with a soft go-live by End of Sep 2025.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5
Product Lines

15+
Countries with Live Installations

1800+
Total Installed Systems Globally

1Million+
Sorts Per Hour

600+
Employees

Long-standing Relationships with industry leaders

Ecom Retail

Flipkart

amazon

Lazada

noon

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

SWIGGY

TITAN

Minister's Warehouse Limited

News & Articles

FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

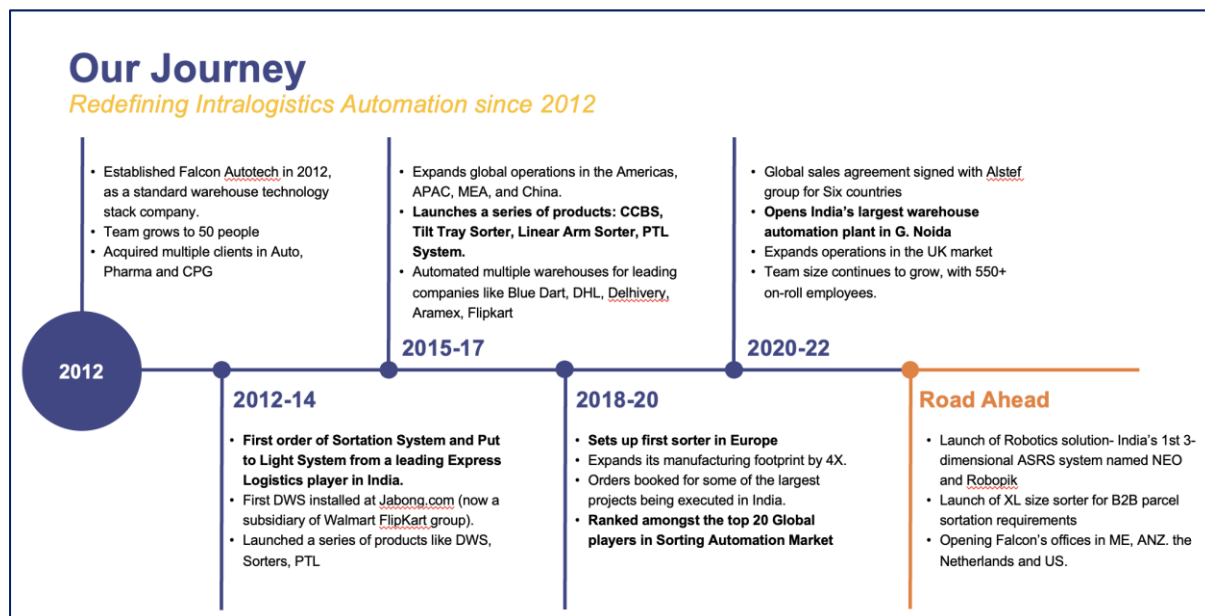
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Pop-up Sorter
- Sweep Sorter
- Pusher Sorter



PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
 - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
 - Mini (600MM)
 - Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

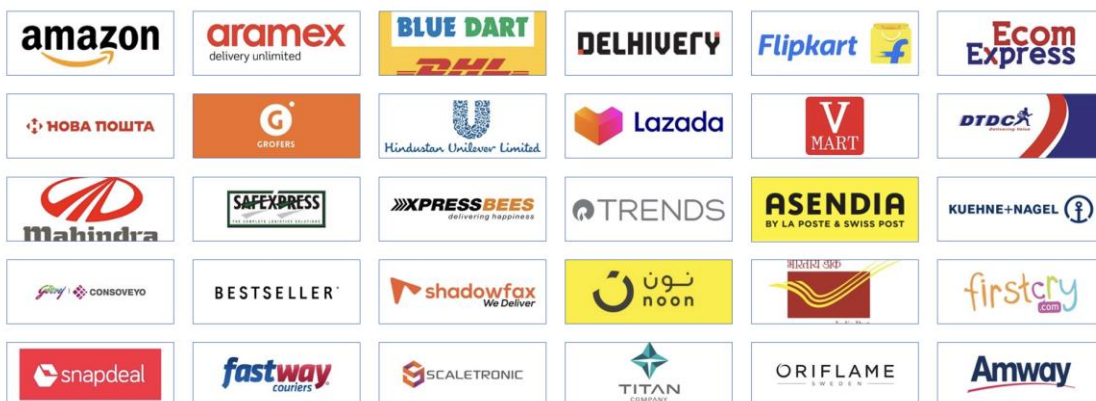


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

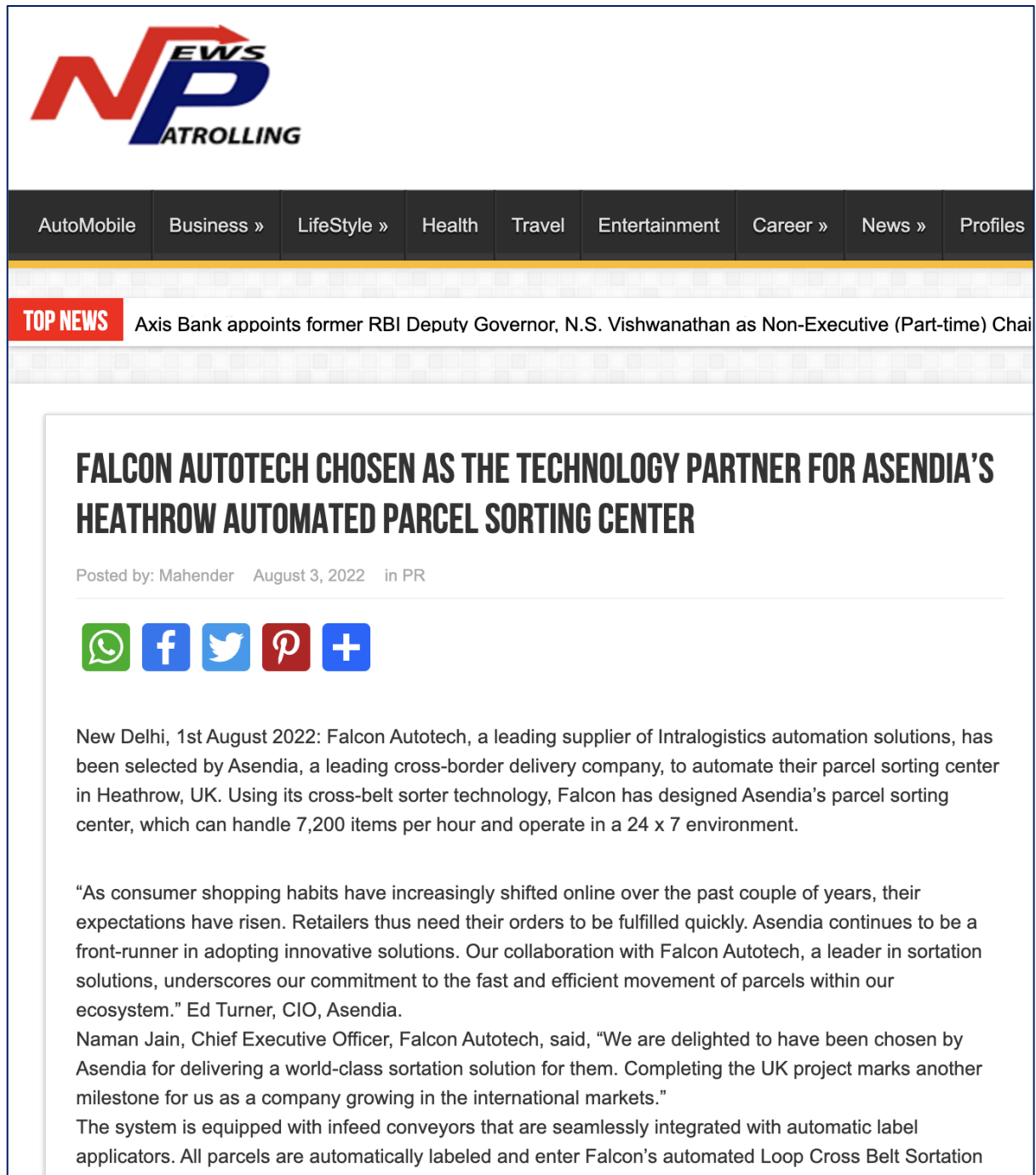


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –



5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

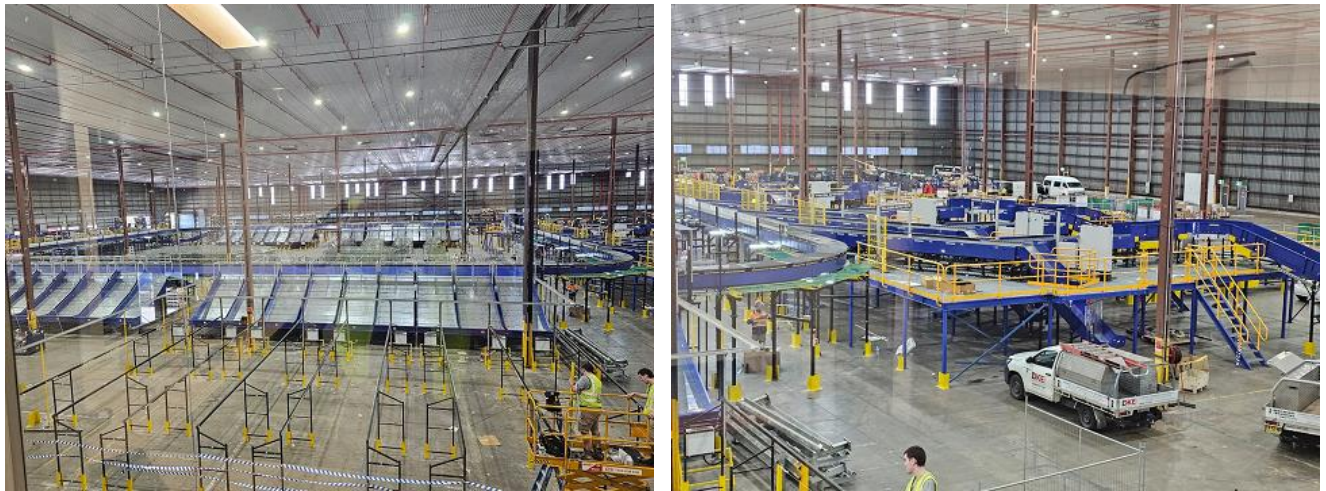
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

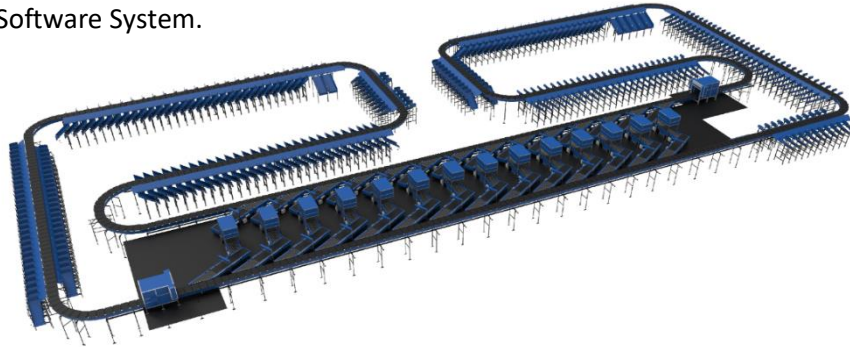
Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.

- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



5.6 Project 6- (Ecommerce Client, India)

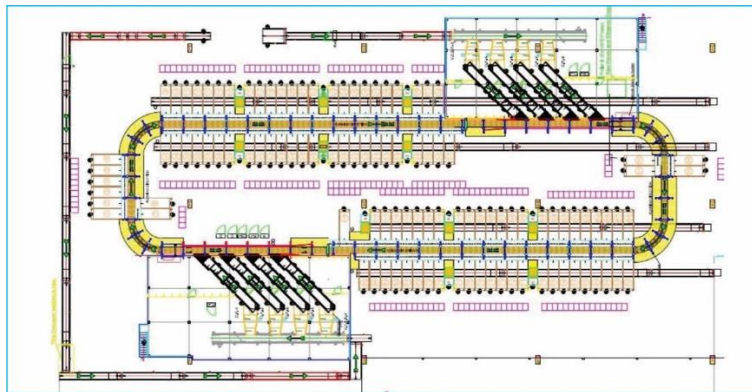
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



6. Handled Parcel Spectrum

As per the parcel spectrum data provided in the RFQ documents, Falcon has studied and analysed the parcel spectrum in detailed.

Falcon purposes to use its “Heavy Dual Belt Cross Belt Sorter” to provide the maximum benefits to DELHIVERY Group in-terms of handling large parcel sizes and weight.

6.1 Parcel size loadable on the sorter

Falcon's heavy duty cross belt sorter has a capability to handle the below mentioned parcel sizes and weight.

Specification	Unit	Value
Max Length	mm	1000
Max Width	mm	800
Max Height	mm	800
Max Weight	Kg	40
Min length	mm	150
Min Width	mm	150
Min Height	mm	50
Min Weight	Kg	0.2
Dimension Accuracy	mm	+/- 10 mm on Standard Box
Weighing Accuracy	kg	+0.060 kg or 0.5% of Parcel weight whichever is higher

Note- Dimension will only be captured for parcels > 50 mm in height.

6.2 Parcels to be loaded on Sorter shall have the following characteristics:

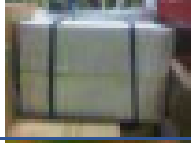
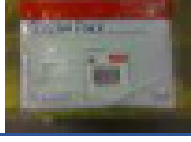
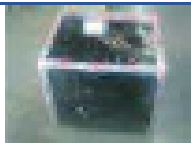
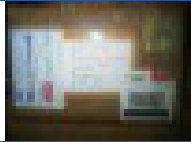
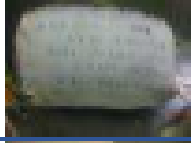
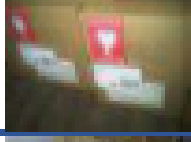
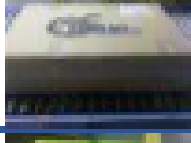
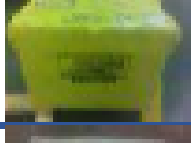
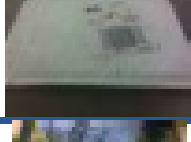
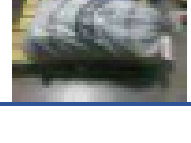
- Centre of Gravity of item must not move during conveyance or sorting.
- Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
- Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
- Plastic ropes shall be perfectly adherent to the surface of the package.
- All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
- Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
- Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
- The parcels must have at least one flat and regular surface providing enough stability during conveyance.
- All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
- All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.4 Pictures of Sortable Parcels

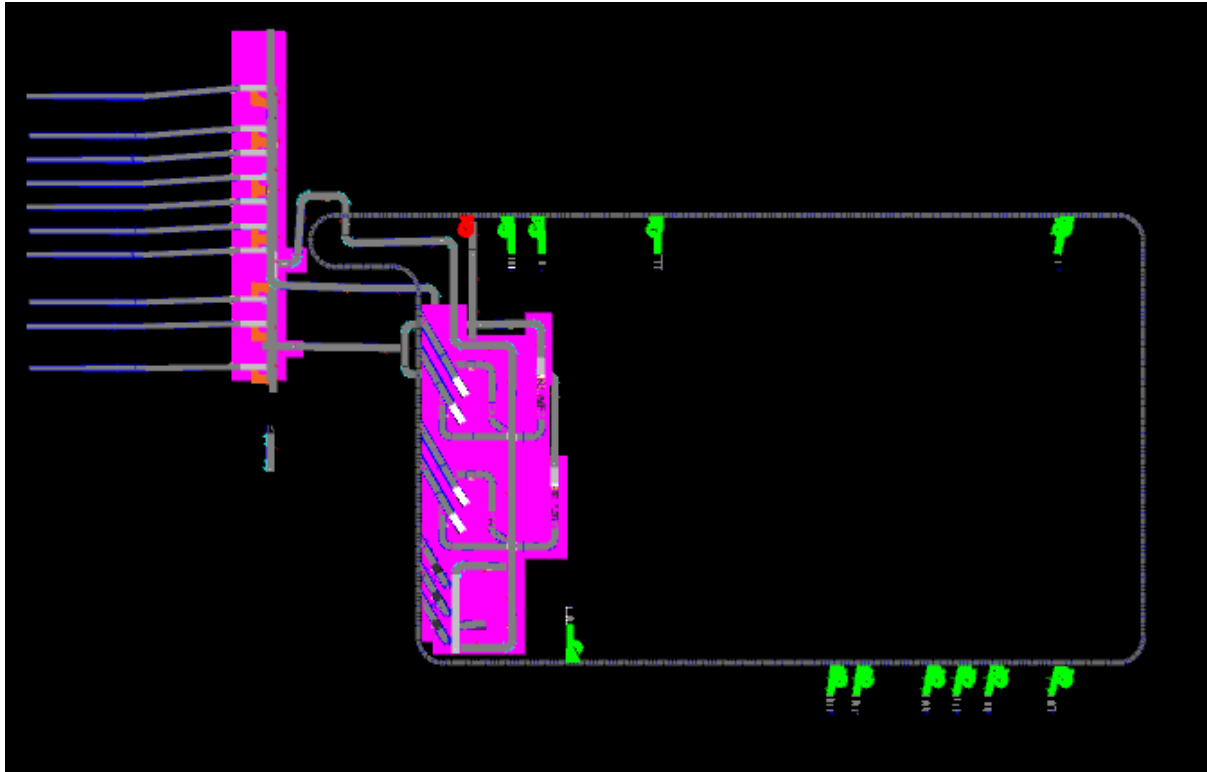
PARCEL TYPE	DESCRIPTION
	Strongly Strapped A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item A cardboard parcel coated in a packing paper.
	Cloth Covered Item Often import parcels that are of irregular shape.
	Wine Box A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box A cardboard parcel is often the norm.
	Totes A large plastic, lidded box.
	Flats A parcel that often contains documents.
	Poly Bags It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Loop CBS for sorting bags & boxes into destination as per specifications shared by DELHIVERY.

7.2 Summary of the System



The complete parcel processing system includes, in accordance with the specifications provided by DELHIVERY:

1. Infeed System:

- Bags & Boxes are loaded manually on the 10 telescopic belt conveyors, which are further connected to infeed highway conveyors.
- These parcels travel from lower level to Mezzanine level and goes through aligning conveyors present before the auto-induct lines.
- 10 manual segregation stations with sliding chutes are positioned strategically on the highway line where operators will segregate bags and transfer them to the Bag Conveyor running below to the Box highway conveyor.
- Boxes on the Common Highway Conveyor will move to the singulators where boxes coming in bulk will be singulated automatically.

- Boxes will move to auto induct feedline (2 set on lower deck) through which the boxes will induct on the sorter.
- Bags are moved to dump VDS loop, from VDS loop operator will pickup the bags and scanned it manually on manual induct lines.

2. **Feedline:** In the proposed solution two types of feedlines are planned:

- a. **Auto Induct:** 2 Set of auto inducts are planned to directly induct the boxes on the CBS Loop
- b. **Manual Induct:** 3 Set of Manual Inducts are planned which will induct bags on the CBS loop.
- c.

3. **Loop CBS:** - Once the parcels are on main loop, the CBS sorts the parcels into respective chutes using data from Delhivery Sorting logic.

4. **Sorting Output:** - The sorted shipments are discharged into below types of chutes.

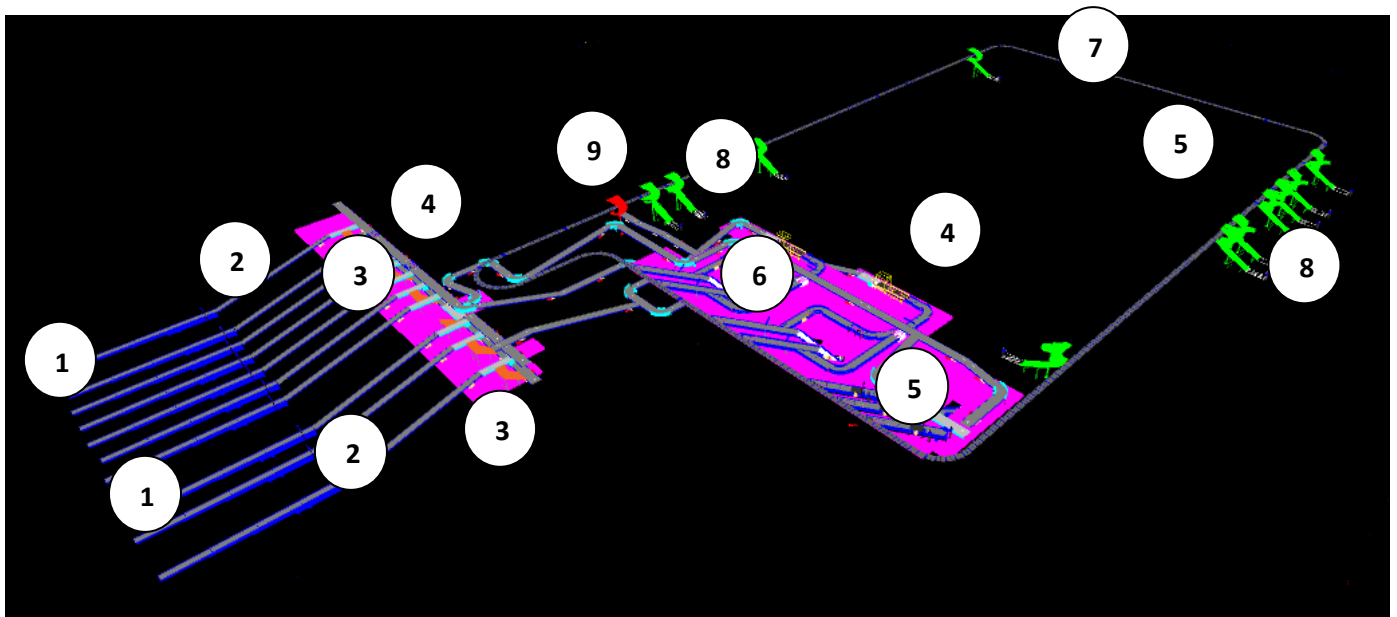
- a. **Outbound Chute**
- b. **Staging Chute**
- c. **Rejection Chute**
- d. **DBO Chute**
- e. **Mixed Bag Chute**

7.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Good Parcels conveying quality for the entire range of conveyable Parcels with a sorting conveyor speed up to 2.3 m/s.
3. Low occupancy of floor space in the building
4. Narrow discharge centers for increased number of splits in limited space
5. FALCON's CBS can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

8. Layout View



Legend

1. Telescopic belt conveyor
2. Infeed Inclined Conveyors
3. Bag Segregation Station
4. 4 – Infeed Highway Conveyor
5. Lower Deck Manual Induct
6. Lower Deck Manual Induct
7. Main Loop
8. Staging Chute
9. DBO Chute

9. Proposed System Capacity Calculations

9.1 Sorter System Capacity

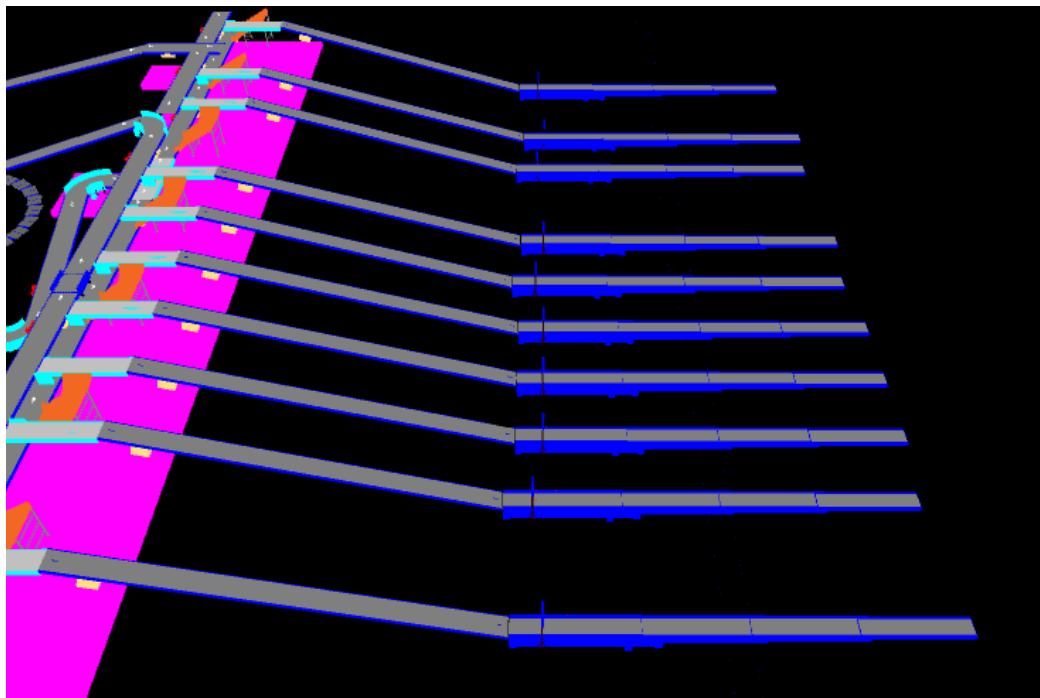
The following table shows the throughput calculation for the sortation system designed based on DELHIVERY's RFP requirements.

SPECIFICATION	VALUE
Sorter Type	Loop CBS
Speed	2.3 m/s
Pitch	1,175 mm
Carrier per hour	7,047 CPH
Belts per hour	14,094 BPH

10. System description

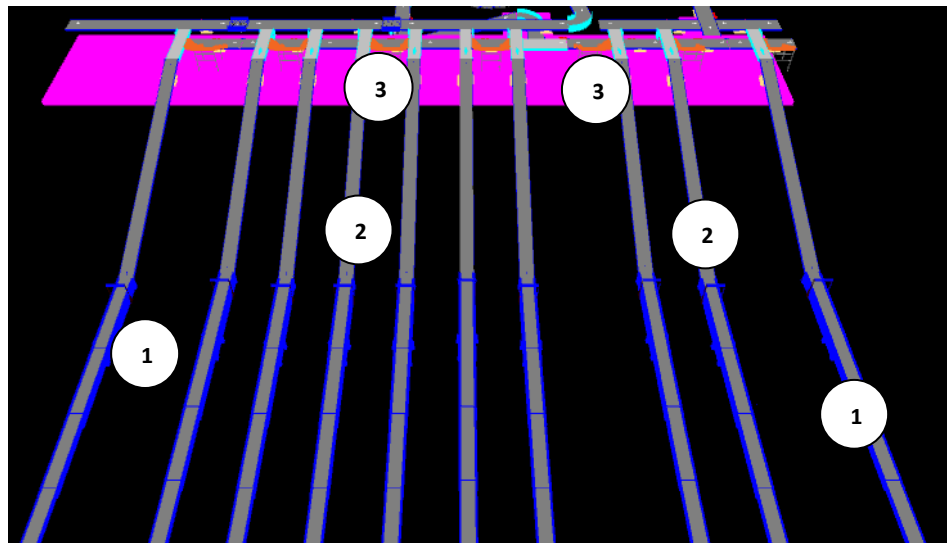
10.1 Unloading Area

Bags & Boxes arriving in Bulk in trucks will be unloaded onto one of the telescopic conveyors present on the dock. On each telescopic conveyor on sensor system is also planned to detect the oversize parcel parcels which will be removed by operator manually.



Truck Unloading Area

10.2 Infeed System



Legend

- 1- Telescopic Belt Conveyor
- 2- Infeed Inclined Conveyor system
- 3- Infeed Highway Conveyor

10.2.1 Telescopic Belt Conveyors

A telescopic belt conveyor, also known as an extendable conveyor or telescoping conveyor, is a type of conveyor system that features a telescoping mechanism to extend or retract the conveyor length as needed.

In the proposed solution, Parcels from the Truck are transferred to infeed conveyors through Telescopic Belt Conveyors in singulated manner.

Specification	UOM	Remark
Manufacturer Name	Name	Guanchao/Equivalent
Suitable Carton Size	MM	Min: 1000x800x800 Max: 100x100x5
Suitable Carton Weight	Kg	Min 0.05, Max 40
Base Length	Mm	7000
Extended Length (Total Length)	Mm	14000
Conveyor Belt Running Speed	m/min	18-30
Motor Rating	Type	0.75 Kw, 415V AC, 50 Hz
Load Capacity	Kg/m	40

10.2.2 Infeed Conveyor System

Infeed conveyor system consists of the following components:

1. Inclined PVC Conveyor
2. Straight PVC Conveyor
3. Modular Conveyor
4. Curve Conveyor
5. Buffer Conveyor
6. Merge Conveyor

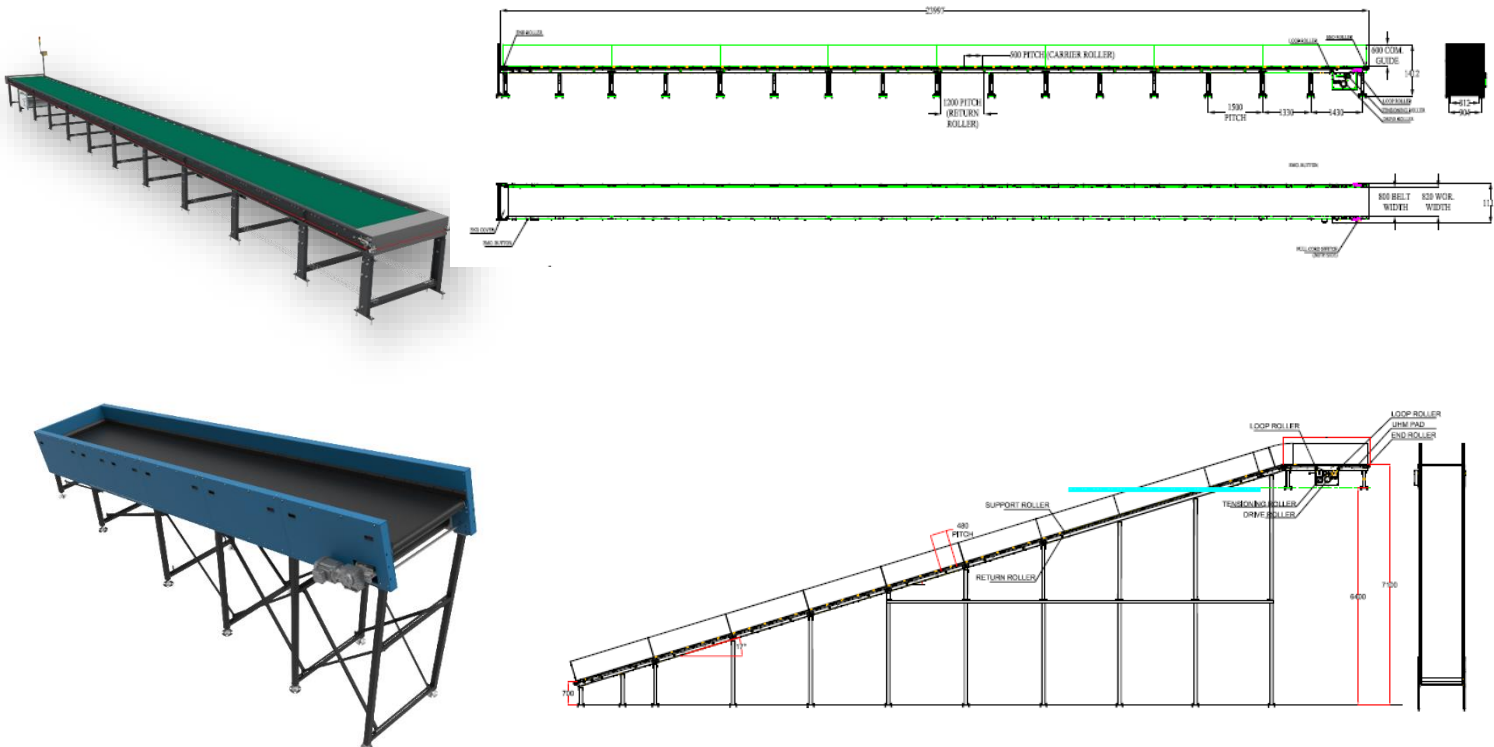
10.2.2.1 Straight and Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



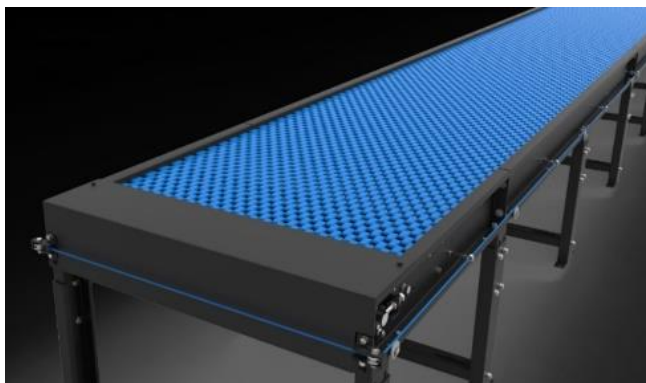
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1000/1200/1400
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord/Similar
Drive	make	Lenze/Siemens/Omron/Allen Bradly/Similar
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

10.2.2.2 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make	Make	Movex
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Wear Strip Material	MOC	UHMW PE1000
Return Roller type	Type	DIA- 50 mm/20303
Return Roller Pitch	mm	800
Driven Shaft	Type	MOC-EN9 Solid Shaft hardened to BHN 240, Size-60SQ MM, Zinc Plated -25 microns, Bearings-UCF-210D1
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

10.2.2.3 Curve Conveyor

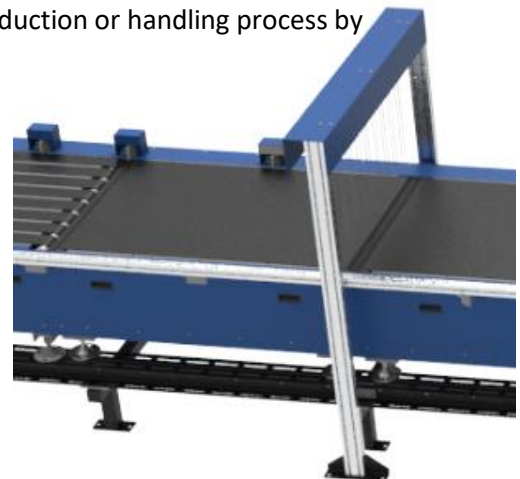
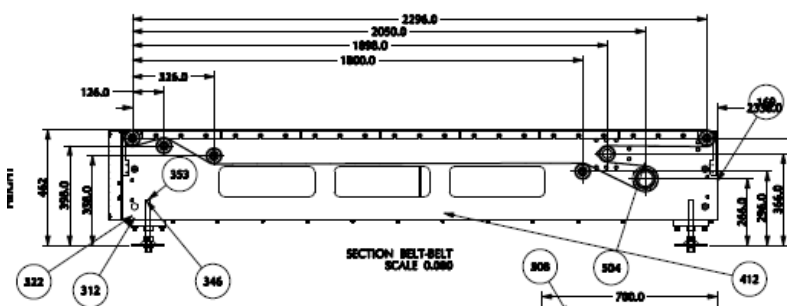
Falcon's curve conveyors are robust and easily maintainable. The uniquely designed curve and belts provides smooth environment to parcels for making the turns. The metal frames of the belts are not deformable to prevent belt misalignment. The belt guide assembly will include removable parts to allow quick replacement in case of damage.



10.2.2.4 Buffer Conveyor

Buffer conveyor is used to temporarily store or hold parcels in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

Buffer conveyor is particularly useful in scenarios where there is a discontinuity or variation in the flow rate of items. They allow for a smooth and continuous production or handling process by absorbing temporary surges or gaps in the flow.

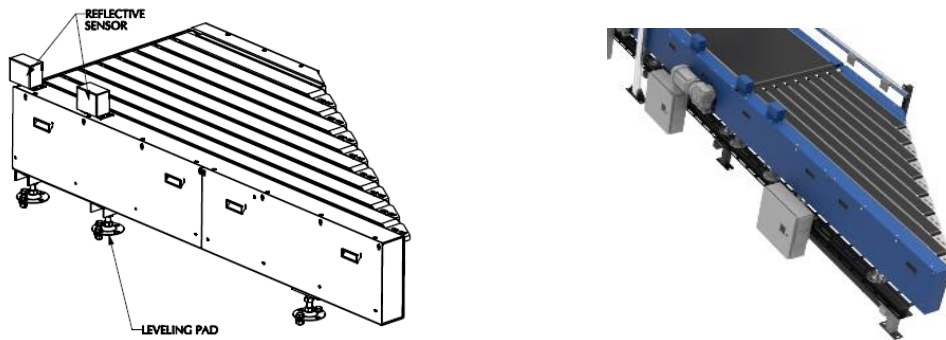


10.2.2.5 Belt Merge

This infeed belt conveyor can be used as a merging means with an angle of 30°, 45° and 60° in the same plane. It can be used, for example, as a follow-up conveyor to a sorter or as a feed unit in a collection circuit.

The nominal angle of this conveyor is achieved by belt modules of different lengths with self-centring belt belts. The optimum belt tension can be set individually for each belt band. Therefore, it is suitable for applications where a high transport speed is required.

The design of the guide rollers allows low-friction operation and low energy consumption; in addition, wear is reduced to a minimum.



10.2.2.6 Aligning Conveyor



Powered Aligning Conveyors are meant for aligning the parcels for ease and efficient induction on cross belt carrier. Aligning conveyors typically employ various mechanisms or techniques to achieve proper alignment.

By incorporating these alignment mechanisms, an aligning conveyor improves efficiency, reduces product damage, and enhances the overall accuracy of the production or handling process.

10.2.3 Conveyor BOQ

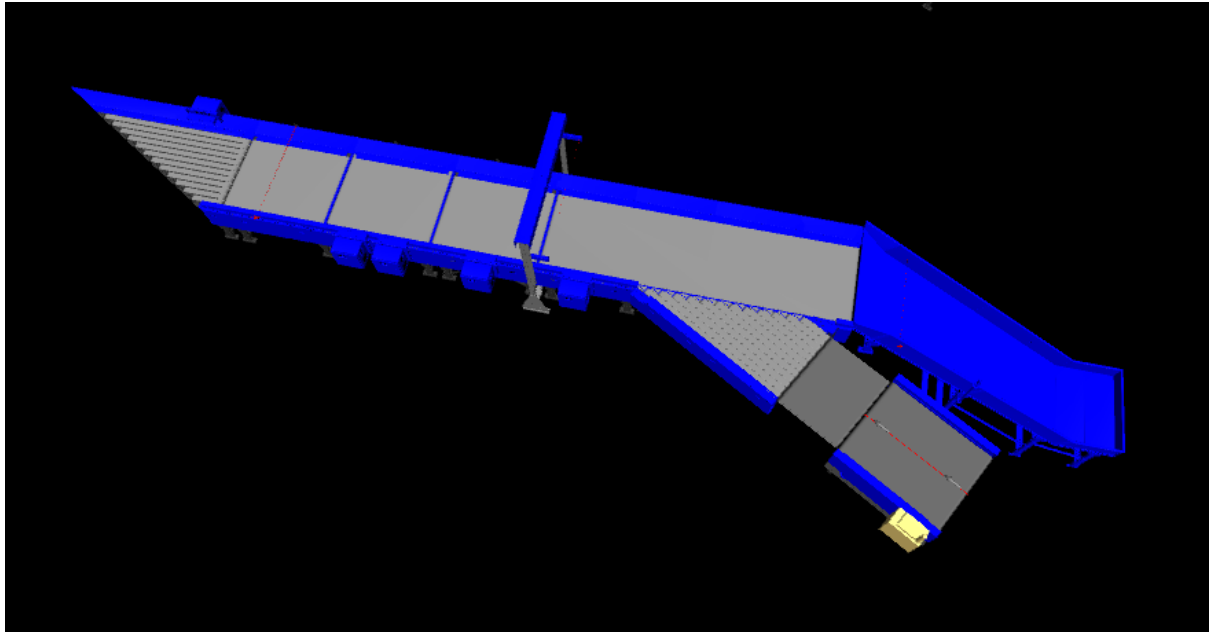
S No.	Name	Angle	EL_1	EL_2	Conveyor Length (m)	Conveyor width (mm)	Set
1	PVC Conveyor	0	4045	4045	2.00	1210	10
2	PVC Conveyor	0	4045	4045	1.95	1210	2
3	PVC Conveyor	0	4180	4180	1.85	1410	1
4	PVC Conveyor	0	1500	1560	4.14	1410	1
5	PVC Conveyor	0	4045	4045	4.32	1410	1
6	PVC Conveyor	0	1500	1500	4.48	1410	1
7	PVC Conveyor	-13	5480	4350	4.80	1410	1
8	PVC Conveyor	0	1500	1500	4.80	1410	1
9	PVC Conveyor	0	3500	3500	4.89	1410	1
10	PVC Conveyor	0	6130	6130	5.28	1410	1
11	PVC Conveyor	0	4180	4180	5.28	1410	1
12	PVC Conveyor	0	1500	1500	5.28	1410	1
13	PVC Conveyor	0	4180	4180	5.76	1410	1
14	PVC Conveyor	0	4180	4180	5.76	1410	1
15	PVC Conveyor	0	6070	6070	5.76	1410	1
16	PVC Conveyor	0	1500	1500	5.76	1410	1
17	PVC Conveyor	0	4180	4180	6.72	1410	1
18	PVC Conveyor	0	4000	4000	7.20	1410	1
19	PVC Conveyor	0	4350	4350	7.44	1410	1
20	PVC Conveyor	7	3430	4350	7.68	1410	1
21	PVC Conveyor	13	4000	5800	7.68	1410	1
22	PVC Conveyor	13	3900	5650	7.68	1410	1
23	PVC Conveyor	0	4180	4180	8.16	1410	1
24	PVC Conveyor	0	4180	4180	8.16	1410	1
25	PVC Conveyor	0	4250	4250	8.16	1410	1
26	PVC Conveyor	0	4180	4180	8.64	1410	1
27	PVC Conveyor	0	6130	6130	8.64	1410	1
28	PVC Conveyor	0	1500	1500	9.27	1410	1
29	PVC Conveyor	-16	4180	1500	9.60	1410	1
30	PVC Conveyor	15	1430	4130	10.56	1410	1
31	PVC Conveyor	14	1430	4045	10.56	1410	1
32	PVC Conveyor	0	1500	1500	11.04	1410	1
33	PVC Conveyor	0	6130	6130	11.52	1410	1
34	PVC Conveyor	0	1500	1500	11.52	1410	1
35	PVC Conveyor	0	6130	6130	13.92	1410	1
36	PVC Conveyor	0	1500	1500	14.40	1410	1
37	PVC Conveyor	0	6130	6130	14.40	1410	1
38	PVC Conveyor	-18	6130	1615	14.40	1410	1
39	PVC Conveyor	0	4000	4000	15.36	1410	1

40	PVC Conveyor	9	1400	4000	16.80	1410	1
41	PVC Conveyor	-14	6000	1560	18.24	1410	1
42	PVC Conveyor	0	5550	5550	18.24	1410	1
43	PVC Conveyor	0	5550	5550	19.68	1410	1
44	PVC Conveyor	16	800	6270	19.68	1010	1
45	PVC Conveyor	16.4	800	6270	19.20	1010	1
46	PVC Conveyor	16	800	6270	19.68	1010	1
47	PVC Conveyor	16	800	6270	19.68	1010	1
48	PVC Conveyor	16.4	800	6270	19.20	1010	1
49	PVC Conveyor	16	800	6270	19.68	1010	1
50	PVC Conveyor	16.4	800	6270	19.20	1010	1
51	PVC Conveyor	16	800	6270	19.68	1010	1
52	PVC Conveyor	16	800	6270	19.68	1010	1
53	PVC Conveyor	16.4	800	6270	19.20	1010	1
54	PVC Conveyor (Bidirectional)	0	6130	6130	6.72	1410	1
55	Modular Conveyor	0	4045	4045	3.00	1448	1
56	Modular Conveyor	0	4180	4180	4.80	1448	1
57	Modular Conveyor	0	4100	4100	14.10	1448	1
58	Modular Conveyor	0	4045	4045	3.00	1448	1
59	Modular Conveyor	0	6200	6200	4.50	1010	1
60	Modular Conveyor	0	6200	6200	4.50	1010	1
61	Modular Conveyor	0	6200	6200	4.50	1010	1
62	Modular Conveyor	0	6200	6200	4.50	1010	1
63	Modular Conveyor	0	6200	6200	4.50	1010	1
64	Modular Conveyor	0	6200	6200	4.50	1010	1
65	Modular Conveyor	0	6200	6200	4.50	1010	1
66	Modular Conveyor	0	6200	6200	4.50	1010	1
67	Modular Conveyor	0	6200	6200	4.50	1010	1
68	Modular Conveyor	0	6200	6200	4.50	1010	1
69	Aligner Conveyor	0	6130	6130	1.60	1400	2
70	Aligner Conveyor	0	6130	6130	1.60	1400	1
71	Aligner Conveyor	0	4045	4045	2.74	1200	2
72	Aligner Conveyor	0	4045	4045	1.60	1200	2
73	Gravity Roller Conveyor	0	750	650	3.00	1410	5
74	Gravity Roller Conveyor	0	900	700	3.00	1350	11
75	Break Roller						48
76	90 Deg Curve					1400	15
77	60 Deg Curve					1200	2
78	90 Deg Curve					1200	6
79	MLG unit for Swedi						2
80	Angle Merge					1200	2
81	Buffer Conveyor					1200	14
82	Spacing Conveyor					1200	4
83	Singulator Unit						2

10.3 Feedlines

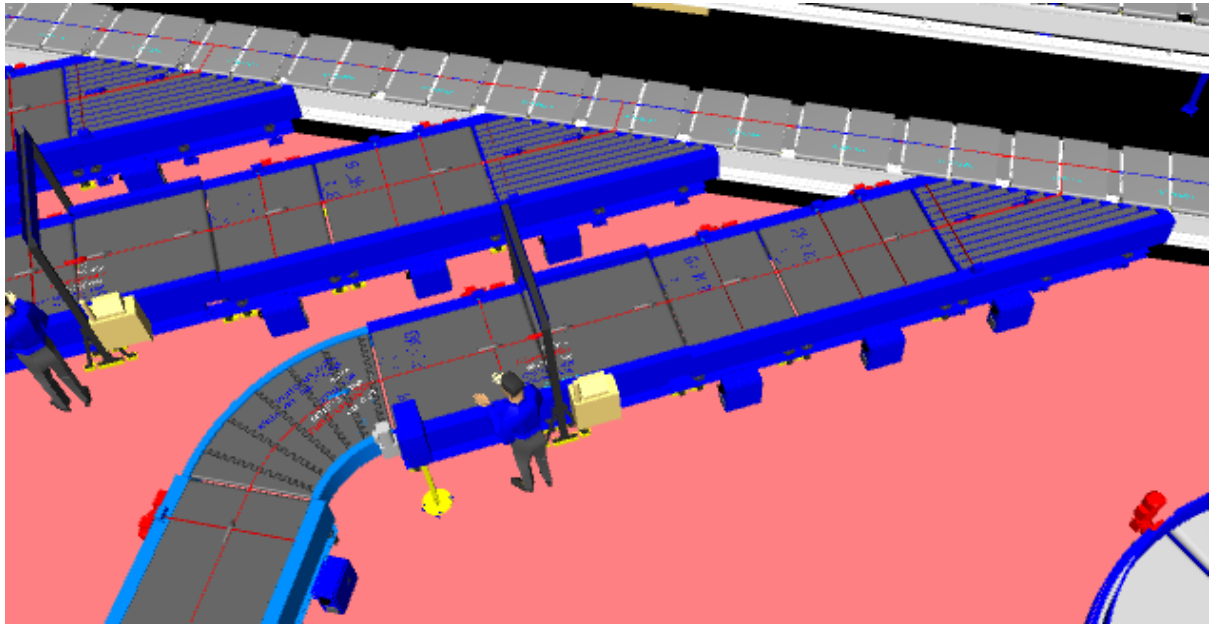
The System has a two type of Feedlines :

- **Auto Induct Feedlines:**



1. Each Auto Induct has set of Eight/Nine Conveyor Modules-
 - a) Spacer Conveyors- 2 Nos.
 - b) Short Buffer conveyor- 3/4 Nos.
 - c) Weighing Conveyor- 1 No.
 - d) Receiving Conveyor- 1 No.
 - e) Angle merge conveyor- 2 No.
2. Rejection chute is provided to take out oversized and overweight parcels from the system.
3. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Parcel on the conveyor and prepare the Cross-Belt Cell for receiving the parcel with high precision.
4. Over the feedline, parcels are evenly spaced and smoothly transferred to the main Loop.

- **Manual Induct Feedlines:**



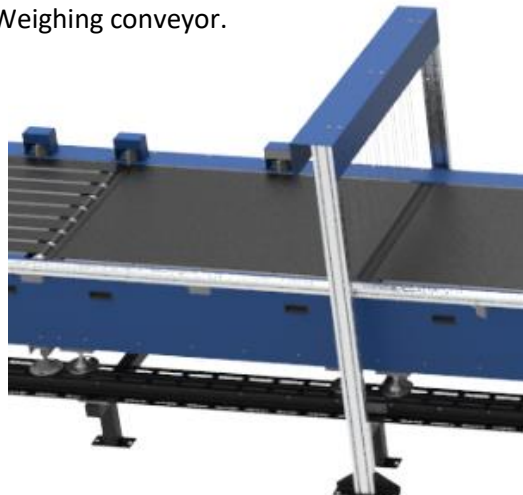
Each Manual Induct has 5 Conveyor Modules-

- f) Loading Conveyor – 1 No
- g) Weighing Conveyor- 1 No
- h) Spacer Conveyors- 2 Nos.
- i) Angle merge conveyor- 1 No.

10.3.1 Spacing Conveyor

A spacing conveyor, also referred to as a gapping conveyor or gap optimizer, is a type of conveyor system used to create and maintain consistent gaps or spacing between items as they move along the conveyor line. Its primary purpose is to regulate the flow and spacing of products to ensure smooth operation and efficient downstream processes.

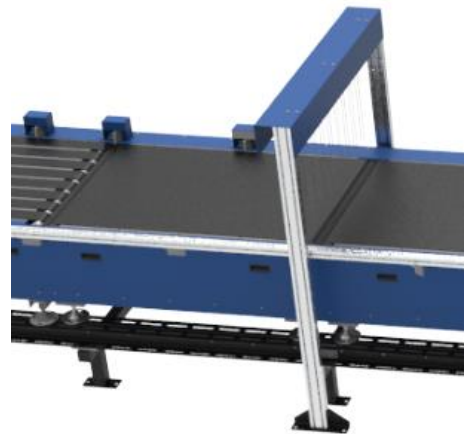
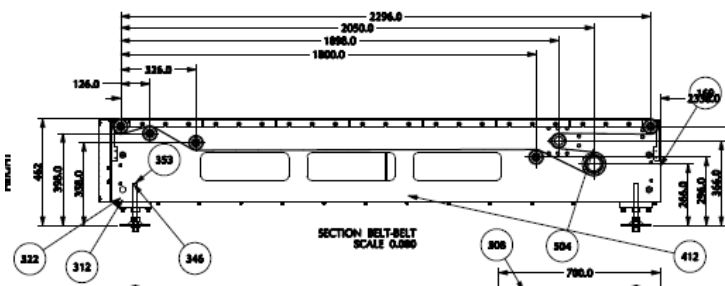
This conveyor is a variable speed special purpose module that creates space between parcels as well as regulates feeding on the Weighing conveyor.



10.3.2 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.



10.3.3 Weighing Conveyor

A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

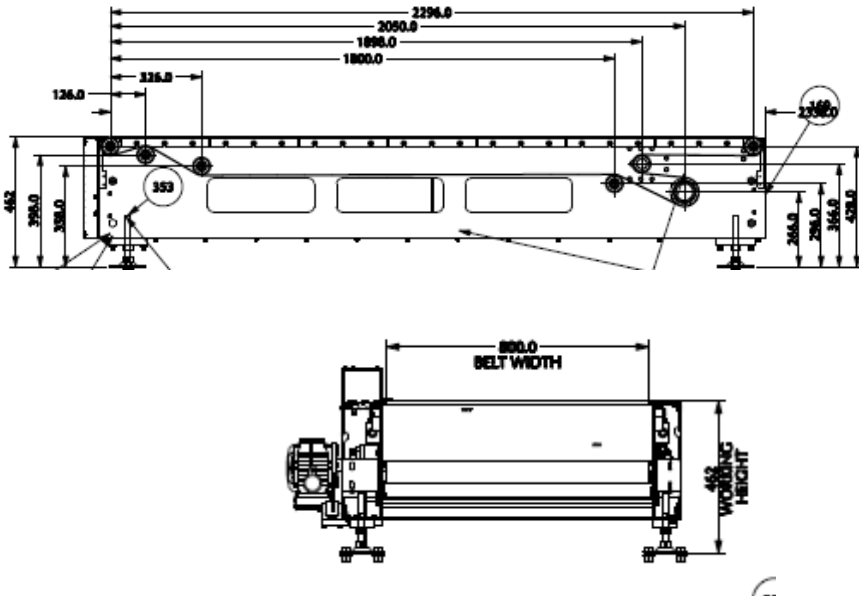
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Parcels.

Make- Bizerba/ Equivalent



10.3.4 Receiving Conveyor

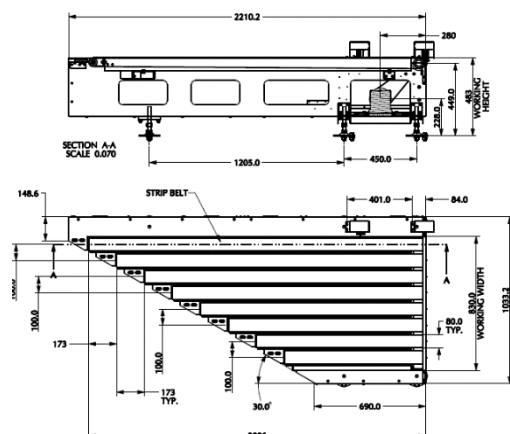
A receiving conveyor, also known as an inlet conveyor or an intake conveyor, is a type of conveyor system used to receive and transfer products from the infeed-line into a processing line. It serves as the initial point of entry where products are collected and conveyed to subsequent stages of the process.



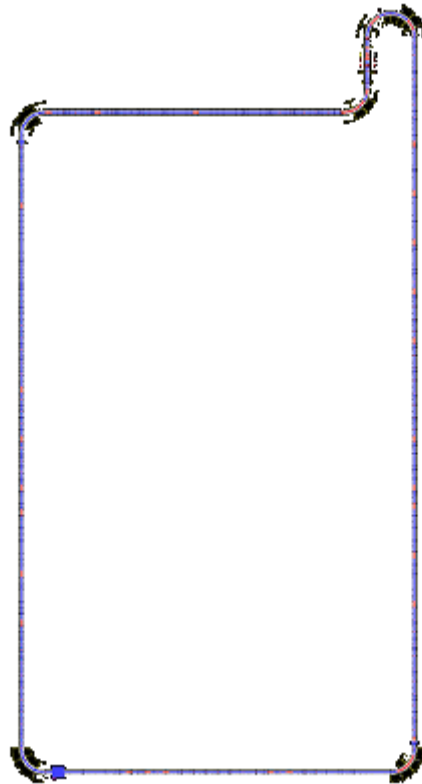
10.3.5 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular Dual Belt high-speed conveyor used for inducing parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.



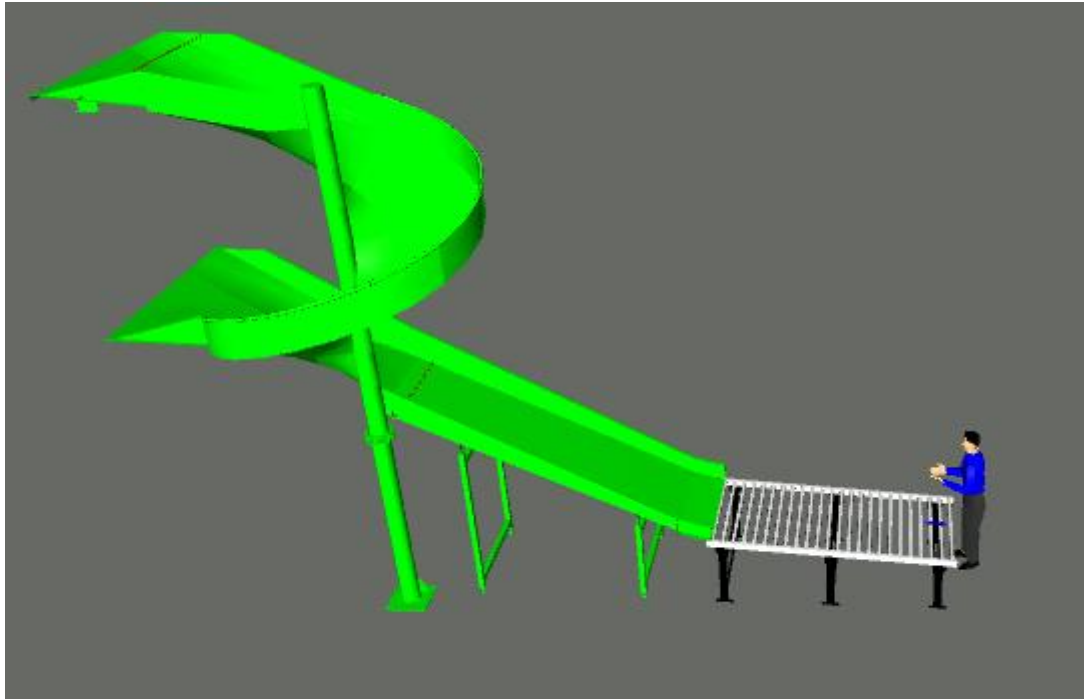
10.4 Main Loop



1. The proposed Single Deck CBS Sorter will be installed on ground level. The upper deck Sorter runs at 6500 mm from ground level and lower deck on 4000 mm from ground level.
2. We have provided loops with dual Belt Carriers with 1175 mm pitch and Belt Size of 495 x 800mm.
3. As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute will actuate and drop the shipment in the assigned chute.

10.5 Sorting Output

10.5.1 Staging Chute



- These are collection-type chutes where bags or boxes accumulate on an idler roller conveyor as sorting continues.
- The user collects the boxes or bags from the roller conveyors.

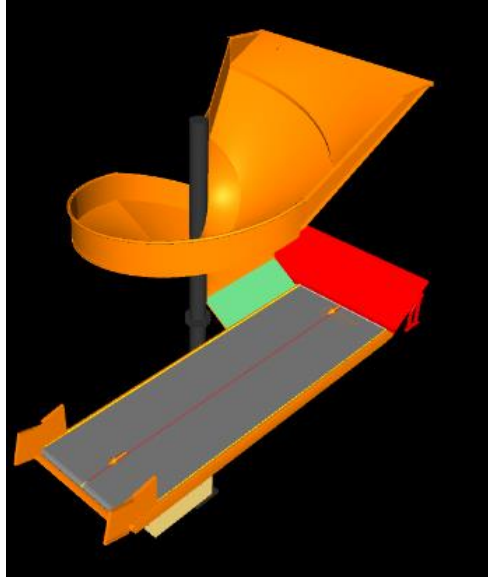
Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Three Color Tower Light

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Three Colour Tower Lamp	To indicate the chute status

10.5.2 DBO Chute

The system is designed with 1 DBO chute. As per the requirement of DELHIVERY, DBO parcels will transfer to the dedicated manual induct for refeeding through after manual scanning.



11. Proposed System Technical Details

11.1 Mechanical equipment

Pos.	Qty.	Description	Value
FM1	1	Infeed System - Powered Belt Conveyors - Modular Belt Conveyor - Merge Conveyor - Aligning Conveyor - Curve Conveyor- 90 Degree - Curve Conveyor- 60 Degree - Buffer Conveyor - Spacing Conveyor - Idler Roller Conveyor - Singulator - Break Roller - Telescopic Belt Conveyor (7+14)	~580 Metres (64 Conveyors) ~70 Metres (14 Conveyors) 2 Nos 7 Nos 21 Nos 2 Nos 14 Nos 4 Nos 48 metres (16 Nos) 2 No 48 Nos 10 Nos .
FM2	3	Manual Induct Feedlines Consists of - Loading Conveyor. - Weighing System - Dynamic Spacing System - Angle merge 2-D Hand Scanners for Manual Scanning	1 Set 1 Set 2 Set 1 Set 1 No.
FM3	4	Auto Induct Feedlines Consists of - Weighing Conveyor - Spacing System - Receiving Conveyor - Buffer Conveyor - Angle Merge	1 No. 2 Set. 1 Nos 3/4 Nos 2 Nos
FM4	1	Falcon Lower Deck Sorter 1 Loop Cross Belt Sorter - Sorter Height - Sorter Length - Sorter Speed - Sorter Drive - Carrier Pitch Including: - Standard Sorter Supports - Empty Carrier Detection System	4000 mm ~444 m ~2.3 m/s Linear Motor 1175 mm

		• Product Centring System • Camera based Barcode reader. • Dimension & 5 side Scanning System • E-Stops • Hooters • Fencing • Netting • Overshoot Assembly	
FM6	1	Sorter Outputs • Staging Chute • DBO Chute • Rejection Chute – Auto Induct • Rejection Chute- Manual Induct • Sliding Chute (Bag Segregation on Infeed System) • Chute Full sensors • Tower Lamp (3 Colour)	11 Nos 1 Nos 4 Nos 1 Nos 7 Nos 24 Nos 17 Nos

11.2 Electrical Equipment

Electricals			
FE1	1	Consists of Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

12. Description of Components of Equipment

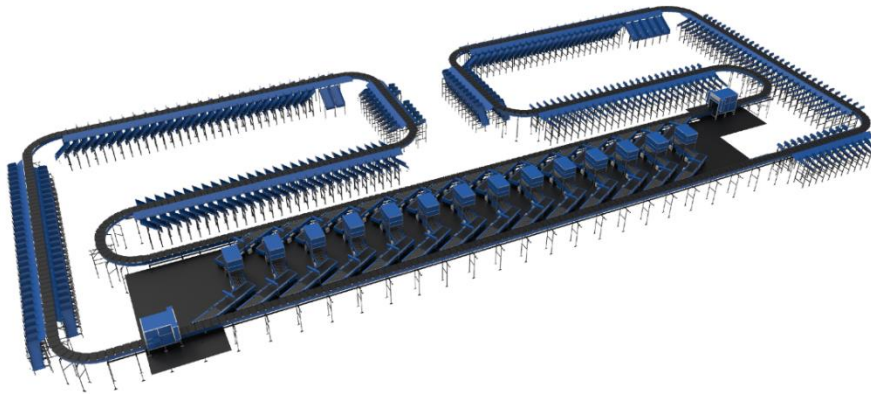
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

12.1 Cross Belt Sorter -

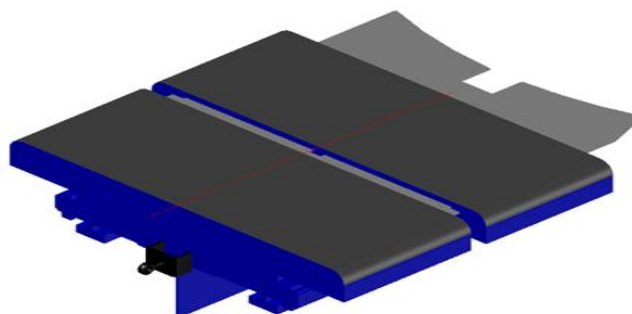
Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



12.2 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners" and non-sortable parcel recirculation.



12.2.1 Servo Roller

High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



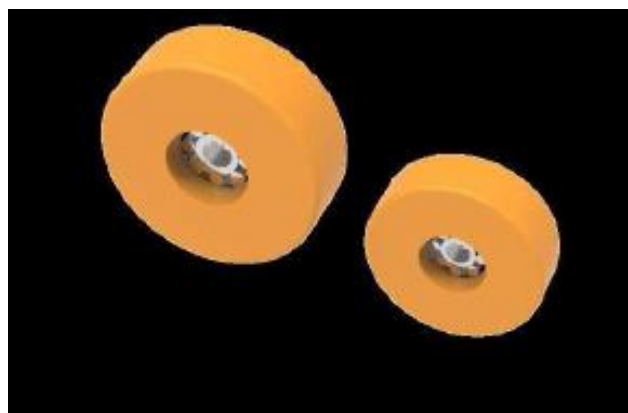
12.2.2 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



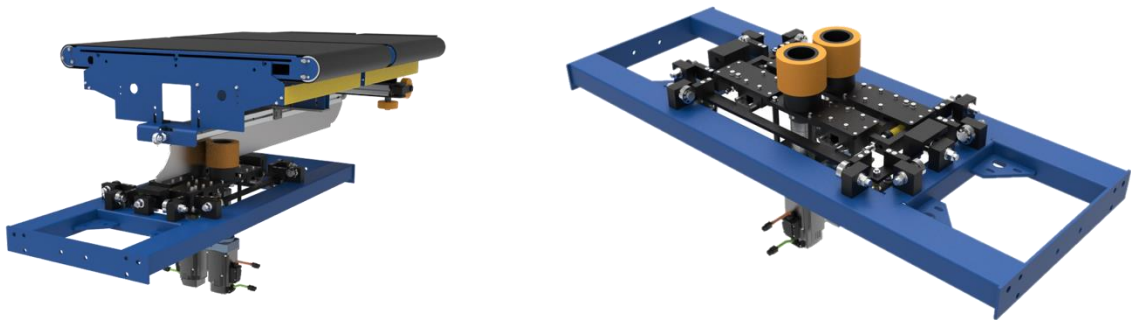
12.2.3 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle.



12.2.4 Friction Wheel Drive

Falcon can provide the indigenously developed Friction Wheel Drive (FWD) to drive the cross belt loop. This driving mechanism operates on the principle of friction, as the name implies. This unit comprises two independent motor-driven wheels that spin in opposite directions synchronised. FWDs are highly energy-efficient drives that promote **sustainability** compared to Linear induction/ synchronous motors.



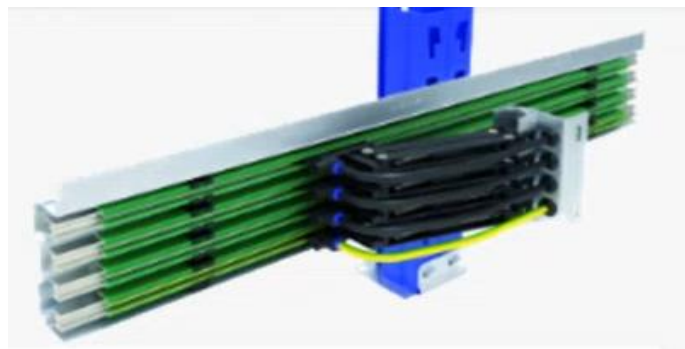
12.2.5 Non-contact based linear motor drive

No Contact based Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.



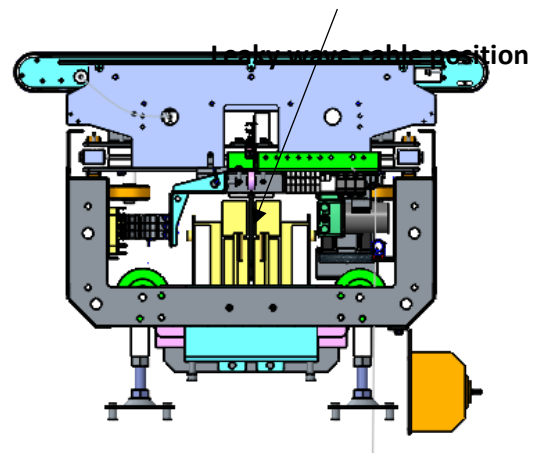
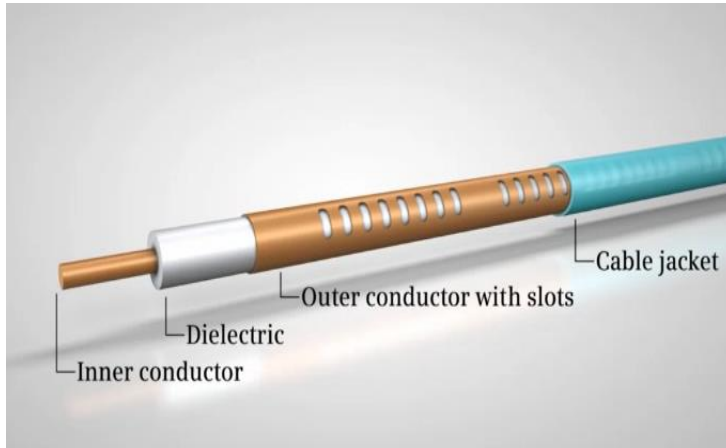
12.2.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



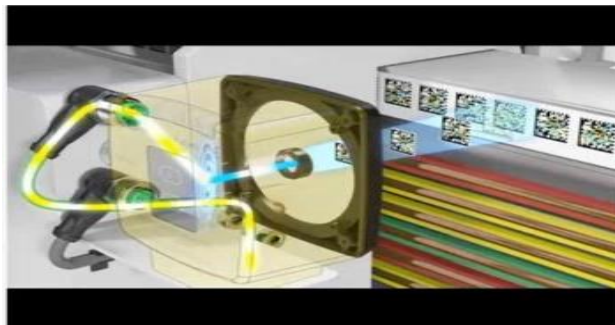
12.2.7 Data Transmission

The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



12.2.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



12.2.9 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2.3 m/s
Sorter Loop Length for lower deck	~444 m
Sorter Actuation Technology	Electric
Chutes	
Staging Chute	11
DBO Chute	1
Height of Sorter Lower Deck	~4000 mm
Track Material to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers	373 Nos. per deck
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during Detailed Design
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	01 Nos per Zone
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

12.3 Scanning & Sensing on Main Sorter Loop

12.3.1 Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time.



12.3.2 Parcel Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



12.3.3 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the parcels on the empty carriers. (Camera based)

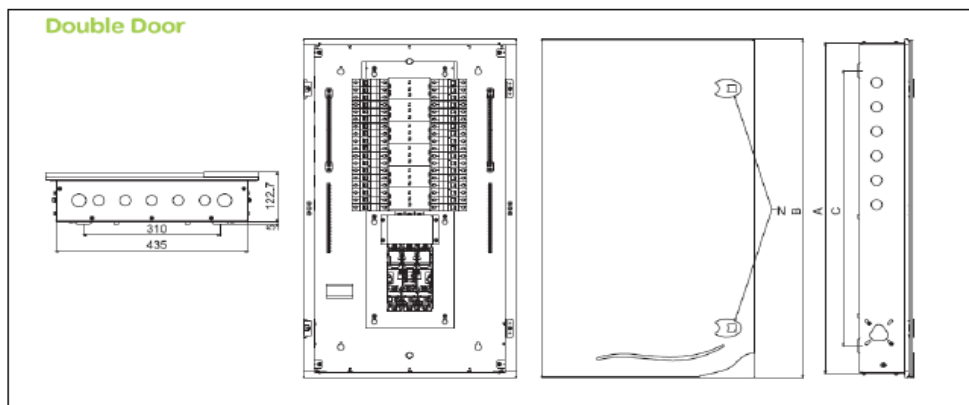


13. Electrical System

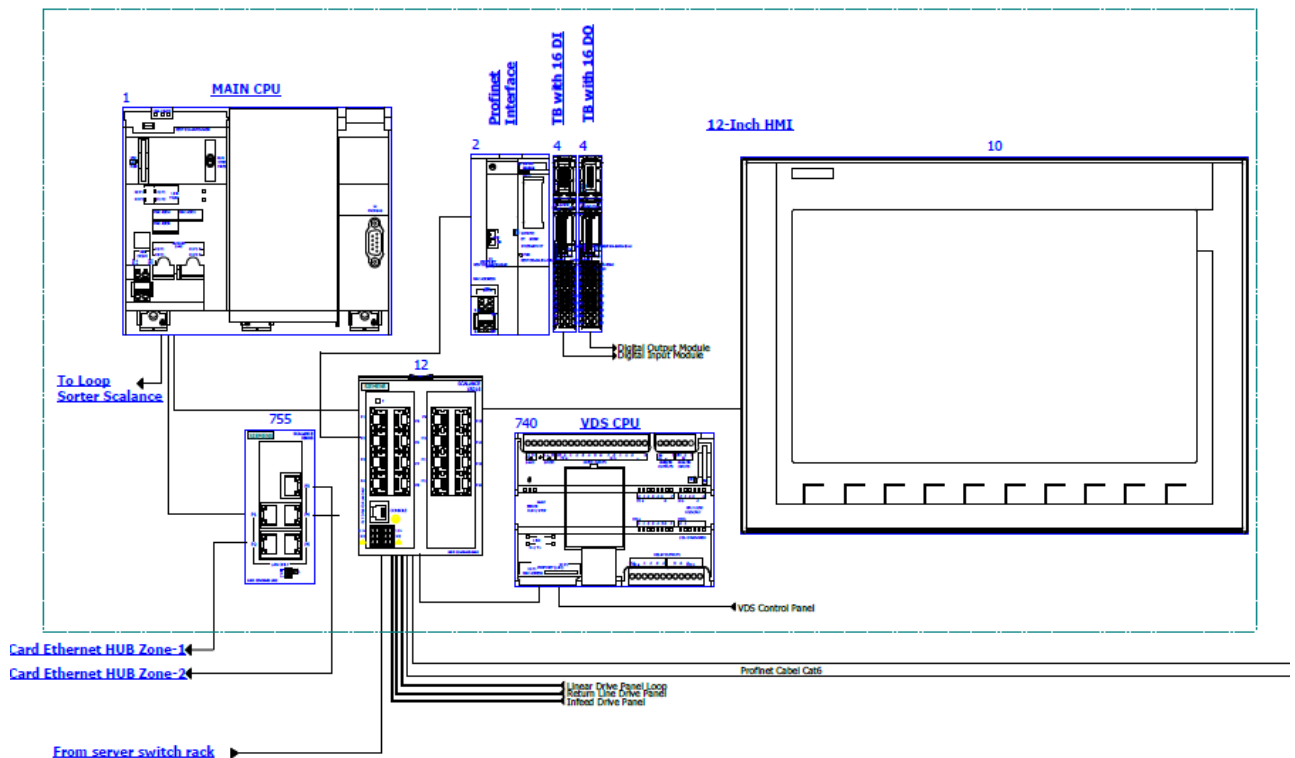
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets. PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

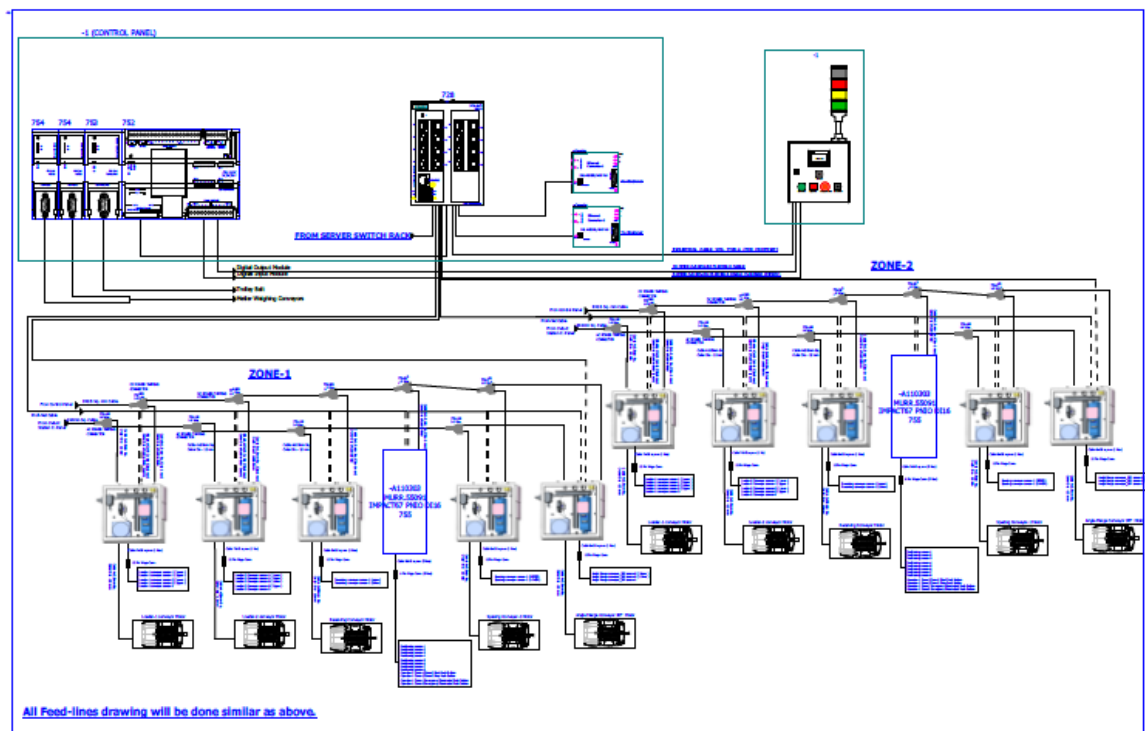
13.1.1 Reference Picture of Power Distribution Panel



13.1.2 Main Control Panel (Reference)



13.1.3 Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter. The engines will be coupled with a converter to improve consumption and reduce the carbon footprint. All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

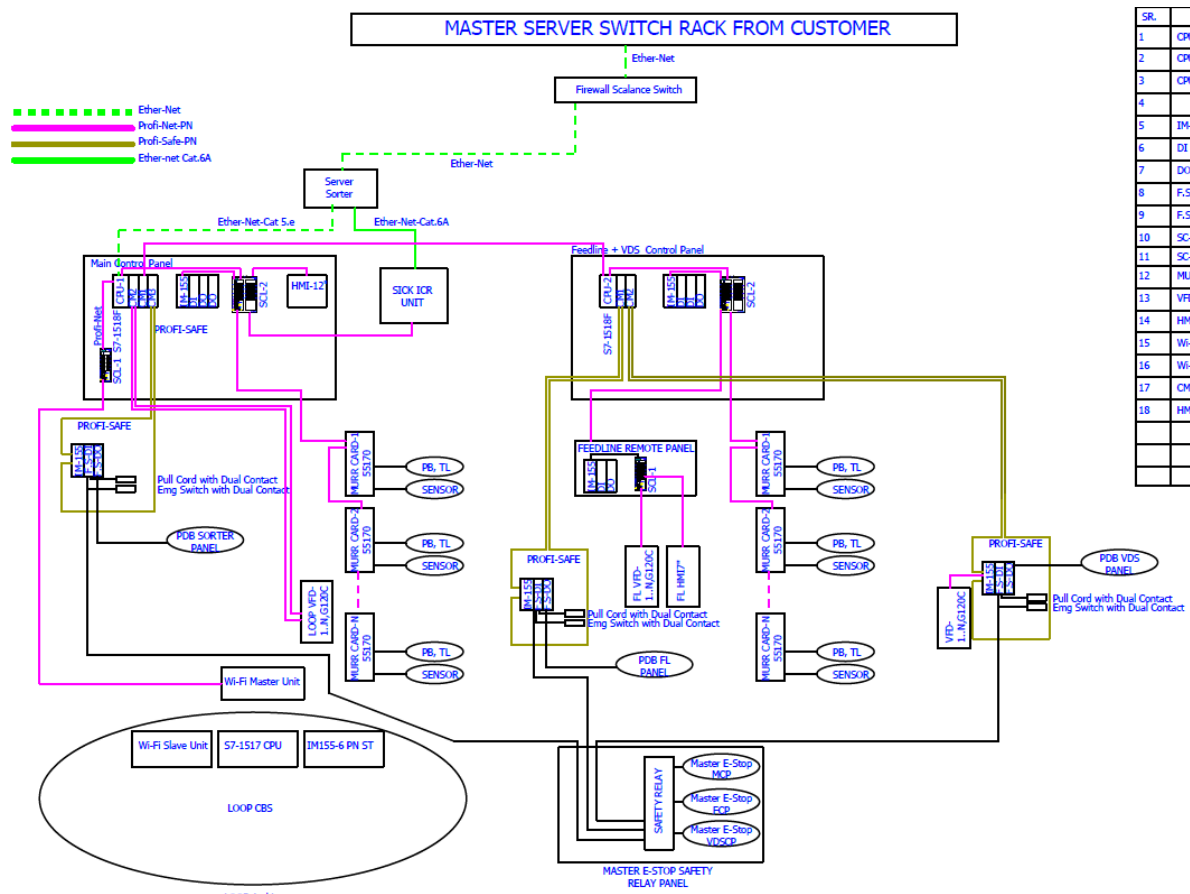
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

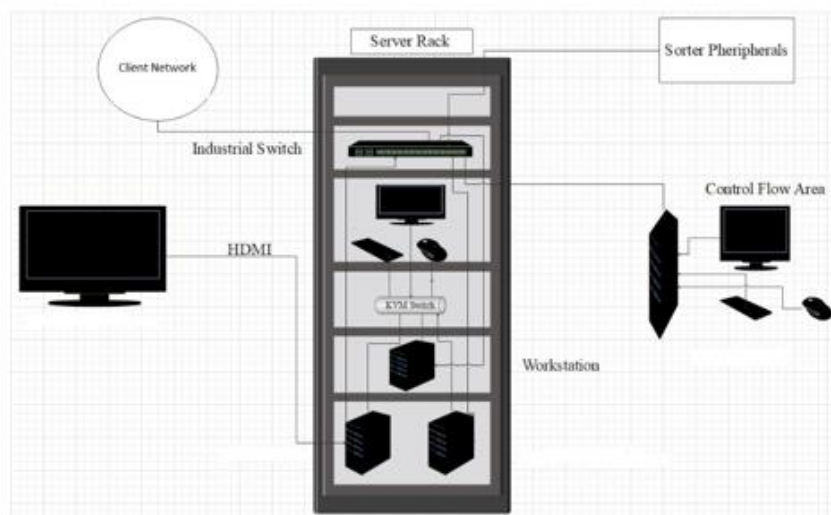
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

14. Controls Architecture (Ref.)



15. IT Architecture (Ref.)

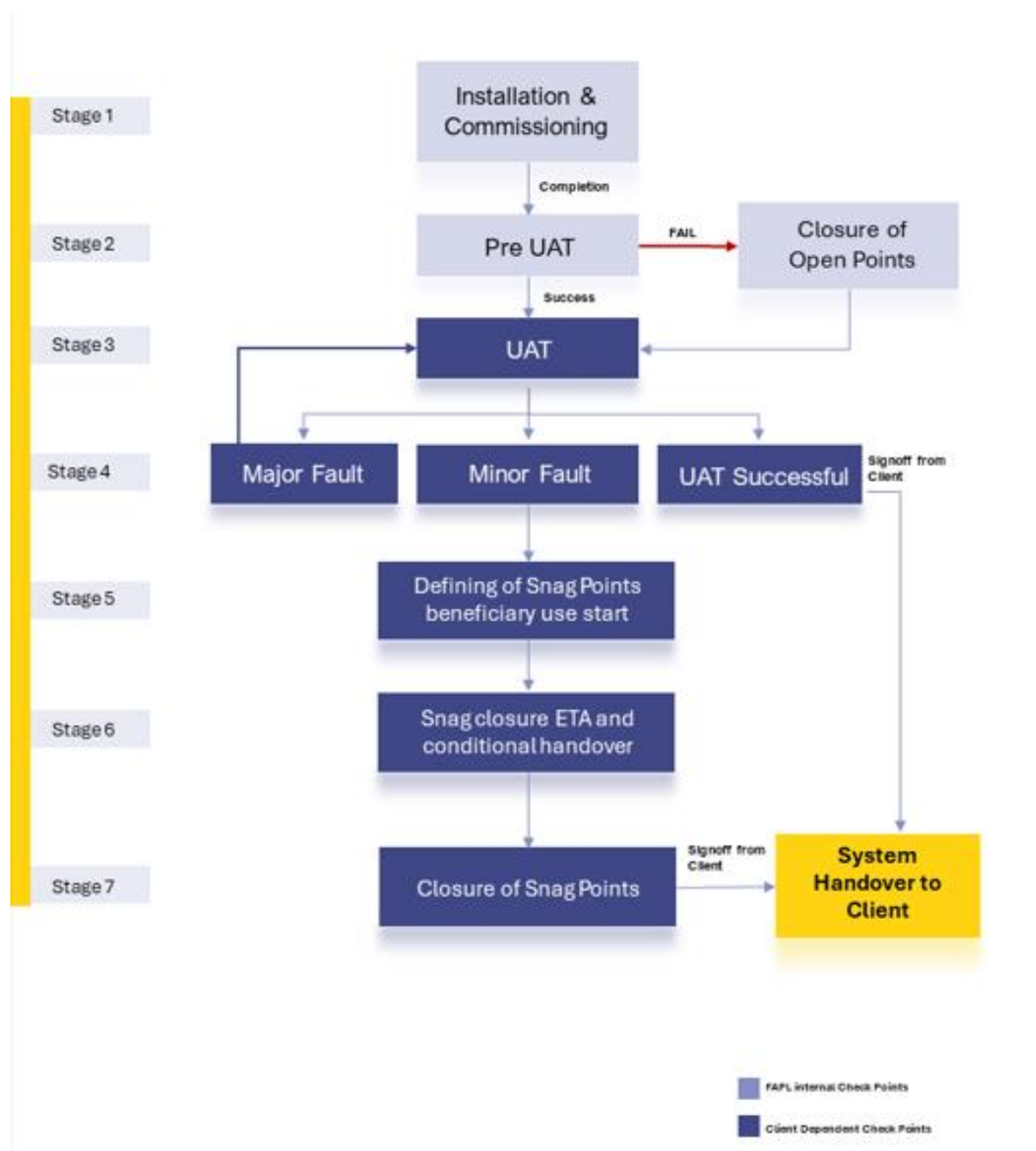


16. Key Components Make

Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze/Equivalent
Volume Scanners	SICK
Encoders	SICK/Falcon/Equivalent
Sensors	Sick/Leuze/ P&F/Equivalent
PLC	Siemens's
Control Panels	Rittal/ BCH
Weighing Scales	Bizerba/Equivalent
VFDs	Siemens/Lenze/ Equivalent
Cables	LAPP /Equivalent
Switch Gear	Schneider/Equivalent
Bearings	NTN/SKF/Equivalent
Data Transmission System	Siemens
Power Transmission Systems	Vahle
HMI's	Siemens/ Equivalent

17. System Handover to Delhivery Team

The system handover will be following the shown below workflow. Each of the stages are defined in the document below.



Installation and Commissioning – Completion of all the activities to get the system up and running.

Pre-UAT – Pre-UAT (User Acceptance Testing) involves a series of preparatory steps and tests conducted before the formal UAT phase begins. This stage ensures that the system is ready for end-user testing and meets the necessary requirements and standards. Below are some key checks of Pre UAT:

- **Physical Verification:** Ensure all functional and non-functional requirements are clearly defined and understood. Verify that all the necessary specifications for the sorting system are documented.
- **Integration Testing:** Conduct integration tests to ensure all components (conveyors, scanners, sorters, control systems, etc.) work together seamlessly. Validate the interfaces between the sorting system and other warehouse management systems (WMS) and databases.
- **Performance Testing:** Test the system's performance with some dummy product to ensure it can handle the expected volume of parcels or items. This test also done to identify and rectify any performance bottlenecks or inefficiencies.
- **Functional Testing:** Execute test cases to verify that the system performs all required functions correctly. Include scenarios for different types of products, error handling, and edge cases.

By thoroughly conducting pre-UAT activities, Falcon team will ensure that the system is stable, efficient, and ready for the formal UAT phase, where end-users can validate its functionality and performance in a real-world scenario.

UAT (User Acceptance Test)

User Acceptance Testing (UAT) for a Cross Belt Sorter System is the final phase in the implementation process where the end users test the system to ensure it meets their requirements and works as expected in a real-world scenario

Falcon will provide the UAT Schedule to Client prior to the commencement of any tests. The UAT Schedule will include the following:

- Test prerequisites.
- Daily plan/schedule
- Personnel responsibilities
- Dashboard data and test loads required.
- Specific test procedures
- Expected test results.

User Acceptance Test Parameters

Below are the parameters/test to be done at site during the UAT :

- Sorter Throughput Test
- Sorter Accuracy Test
- Barcode Reading Test

**SOP for all test will be shared during DAP stage*

.

Note: In conducting this evaluation/UAT, Falcon team adhere to its Standard Operating Procedure

Minor & Major Faults

Any faults encountered during Testing will be categorized by Falcon as either Minor or Major faults as defined below.

- **Minor Faults** are defined as those affecting a limited area or single component that has no impact on the test. Minor Faults will be added to the System Snag point list and will not inhibit the continuation of testing.
- **Major Faults** are defined as those having impact that result in the inability to demonstrate the subject functionality. Major Faults may require re-starting of the test.

System Snag Point

Once the UAT completed, all the minor faults will be considered as snag points. Falcon team will publish Snag Point List information to the client.

System Snag Point list information will include the following:

1. Issue.
2. Date identified.
3. Area and/or unit.
4. Category (mechanical, electrical, or software).
5. Remedy responsibility (responsible person[s] or organization[s]).
6. Target completion date.
7. Snag point completion verification and signoff.

Once the snag point list information's shared with client, client can start the beneficiary use of system and also Client needs to sign off on Start of System Warranty and conditional handover.

Practice for Snag Point Sign Off:

1. Falcon personnel will maintain and distribute the System Sang List throughout UAT.
2. As each issue is corrected, Client will verify the resolution and provide sign-off for that issue on the System Snag Point List.

System Handover Letter

After successful completion of Acceptance Testing or closure of all snags, Falcon will furnish Client a letter, which will address the following:

- The subject System has been installed and accepted by Client.
- The final payment amount that is required and the designated due date.

18. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (**Starts from the date of beneficiary use**) and post completion of warranty, customer can opt for Extended warranty.

Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

19. Commercial Terms

19.1 Price Sheet

S. No	Component	Qty per set	Set	UOM	Total (INR)
1	Loop Cross Belt Sorter with ICR, VMS (Lower Deck)	1	1	Set	₹ 10,17,04,268
2	Conveyors & Swedi	1	1	Set	₹ 9,19,67,818
3	Chutes & Its Accessories	1	1	Set	₹ 1,00,38,862
4	Software Packages & Tool Kit	1	1	Set	₹ 3,00,000
5	Installation & Commissioning	1	1	Set	₹ 61,11,328
6	Packaging & Forwarding	1	1	Set	₹ 20,37,109
7	Project Management and Engineering Charges	1	1	Set	₹ 10,60,797
Total					₹ 21,32,20,183
BCA Discount (5 %)					₹ 1,06,61,009
Final					₹ 20,25,59,174
TBC 10 Nos					₹ 1,44,00,000
Final					₹ 21,69,59,174

The Total Price is to be understood.

- Freight- Excluded
- Taxes: Extra as Applicable
- Including Packing and Forwarding
- Installation and Commissioning: Included

*Material Unloading & Laydown area are in customer scope

19.2 Payment Terms

S.No	Percentage	Stage
1	40%	Along with Purchase Order within 7 days of contract signing
2	50%	Against Delivery of material at site with 100% Taxes on Pro Data Basis
3	10%	Against Installation & Commissioning

**All invoices (if due as per payment terms) should be cleared within 30 days from the date of receipt of undisputed invoice by the buyer*

20. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Construction Power**
- **Leg Guards**
- **Mezzanine**
- **Steel Works- If not specified**
- **Collection Trolleys.**
- **PC for SCADA.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.